

A SHINY APP FOR RIGOROUS AND REPRODUCIBLE MENDELIAN RANDOMIZATION

Eastern South Dakota Research Symposium

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COLLABORATIVE RESEARCH WITH JI HOON PARK



WHAT IS A SNP?

Individual 1

Maternal ...CGATATTCC**T**ATCGAATGTC...

Paternal ...CGATATTCC**C**ATCGAATGTC...

Individual 2

Maternal ...CGATATTCC**C**ATCGAATGTC...

Paternal ...CGATATTCC**C**ATCGAATGTC...

Individual 3

Maternal ...CGATATTCC**T**ATCGAATGTC...

Paternal ...CGATATTCC**T**ATCGAATGTC...

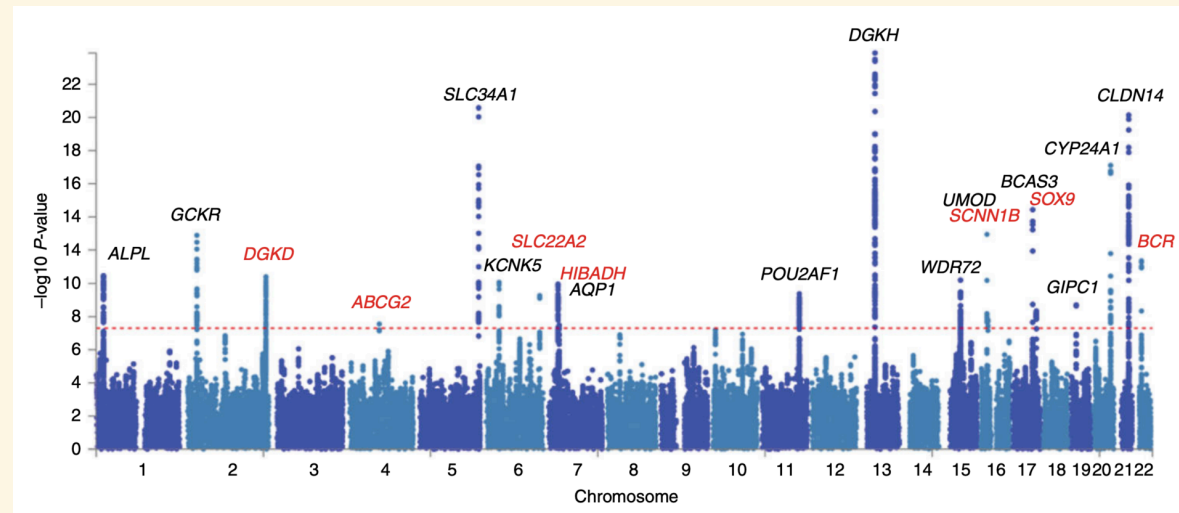
Individual 4

Maternal ...CGATATTCC**C**ATCGAATGTC...

Paternal ...CGATATTCC**T**ATCGAATGTC...

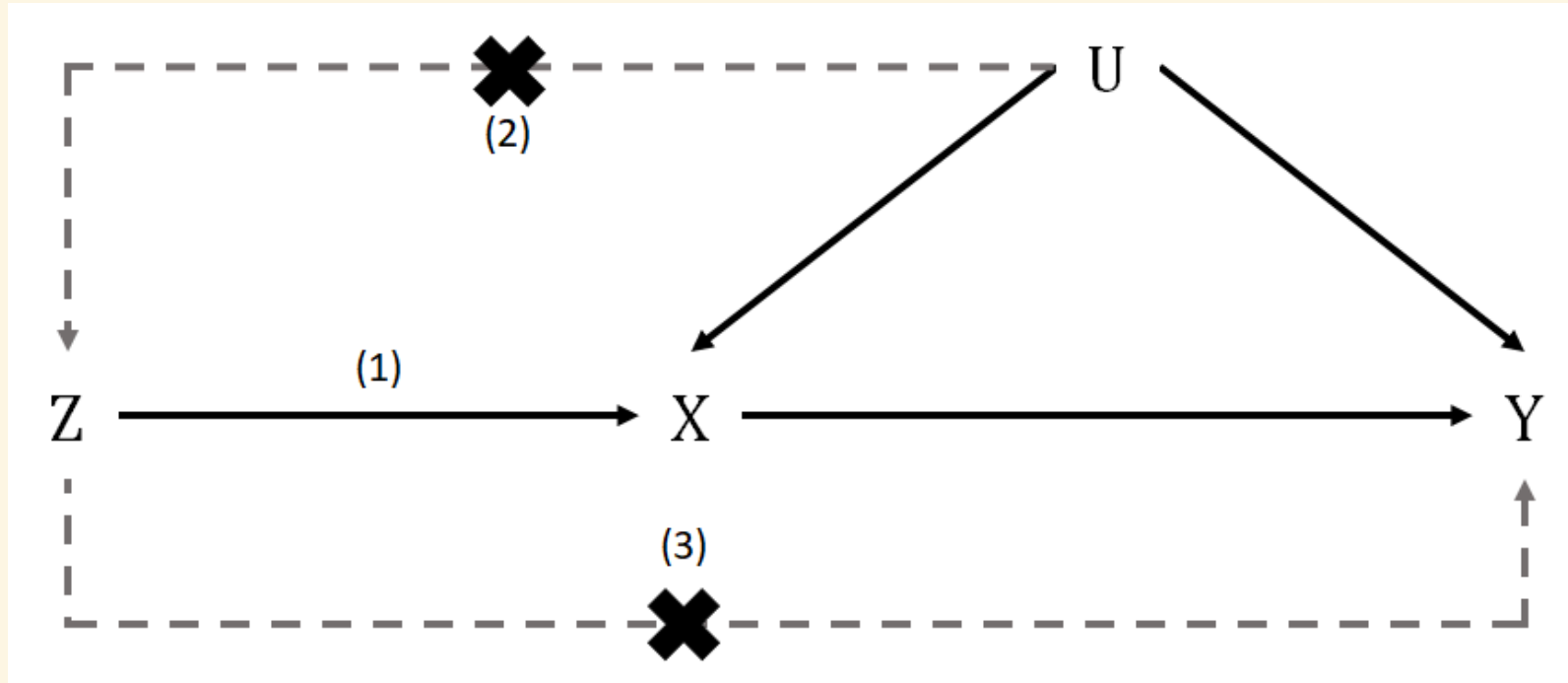
GENOME-WIDE ASSOCIATION STUDIES

- A study to detect genotype-phenotype associations
- Involves thousands of subjects with SNP genotypes and phenotypes
- GWAS results per SNP are publicly shared



GWAS for kidney stone disease

WHAT IS MENDELIAN RANDOMIZATION?



- Instrumental variable analysis to characterize evidence for a causal effect of the “exposure” (X) on the “outcome” (Y)

WHAT IS SHINY?



- An R package that enables users to build an interactive webpage “App”
- App can then be deployed on a website for other scientists to use

MOTIVATION FOR A SHINY APP

- Many recent reports have questioned the scientific rigor of published MR studies
- To promote rigor and reproducibility, users can download R code to perform their MR analyses

SHINY APP DESIGN CONSIDERATIONS

- Interface with openGWAS database
- Require sensitivity analyses to assess MR assumptions
- Users must manually ensure that there is no subject overlap in the two GWAS

DEVELOPMENT VERSION OF THE APP

Trait SelectionInstrumentsOutcome DataMR ResultsScatter Plot

Exposure Trait Keyword:

Outcome Trait Keyword:

Selected Exposure ID: ebi-a-GCST90000066

Selected Outcome ID: ebi-a-GCST009541

Filtered Exposure Trait Rows

Show 10 entries

Search:

	id	trait	coverage	ncase	group_name	year	mr	author	sex	qc_prior_to_upload	pmid	priority	population
20	ebi-a-GCST90000066	Systolic blood pressure			public	2020	1	Surendran P	NA		33230300		European
21	ebi-a-GCST90000065	Pulse pressure			public	2020	1	Surendran P	NA		33230300		European
22	ebi-a-GCST90000063	Diastolic blood pressure			public	2020	1	Surendran P	NA		33230300		European
23	ebi-a-GCST90000062	Systolic blood pressure			public	2020	1	Surendran P	NA		33230300		European
24	ebi-a-GCST90000061	Pulse pressure			public	2020	1	Surendran P	NA		33230300		European
25	ebi-a-GCST90000059	Diastolic blood pressure			public	2020	1	Surendran P	NA		33230300		European
165	ieu-b-38	systolic blood pressure			public	2018	1	Evangelou, E	Males and Females		30224653	0	European
166	ieu-b-39	diastolic blood pressure			public	2018	1	Evangelou, E	Males and Females		30224653	0	European
5	ebi-a-GCST90029010	Diastolic blood pressure			public	2018	1	Loh PR	NA		29892013		European

THANK YOU!